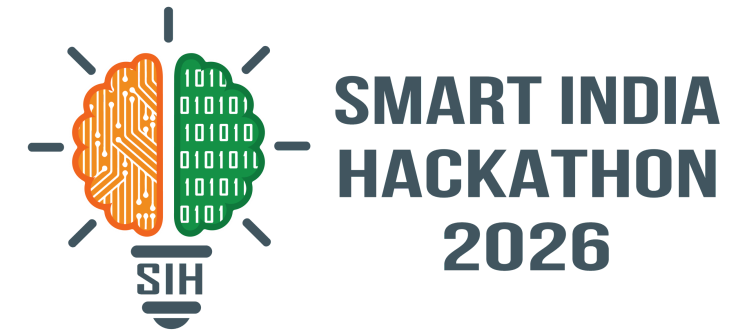
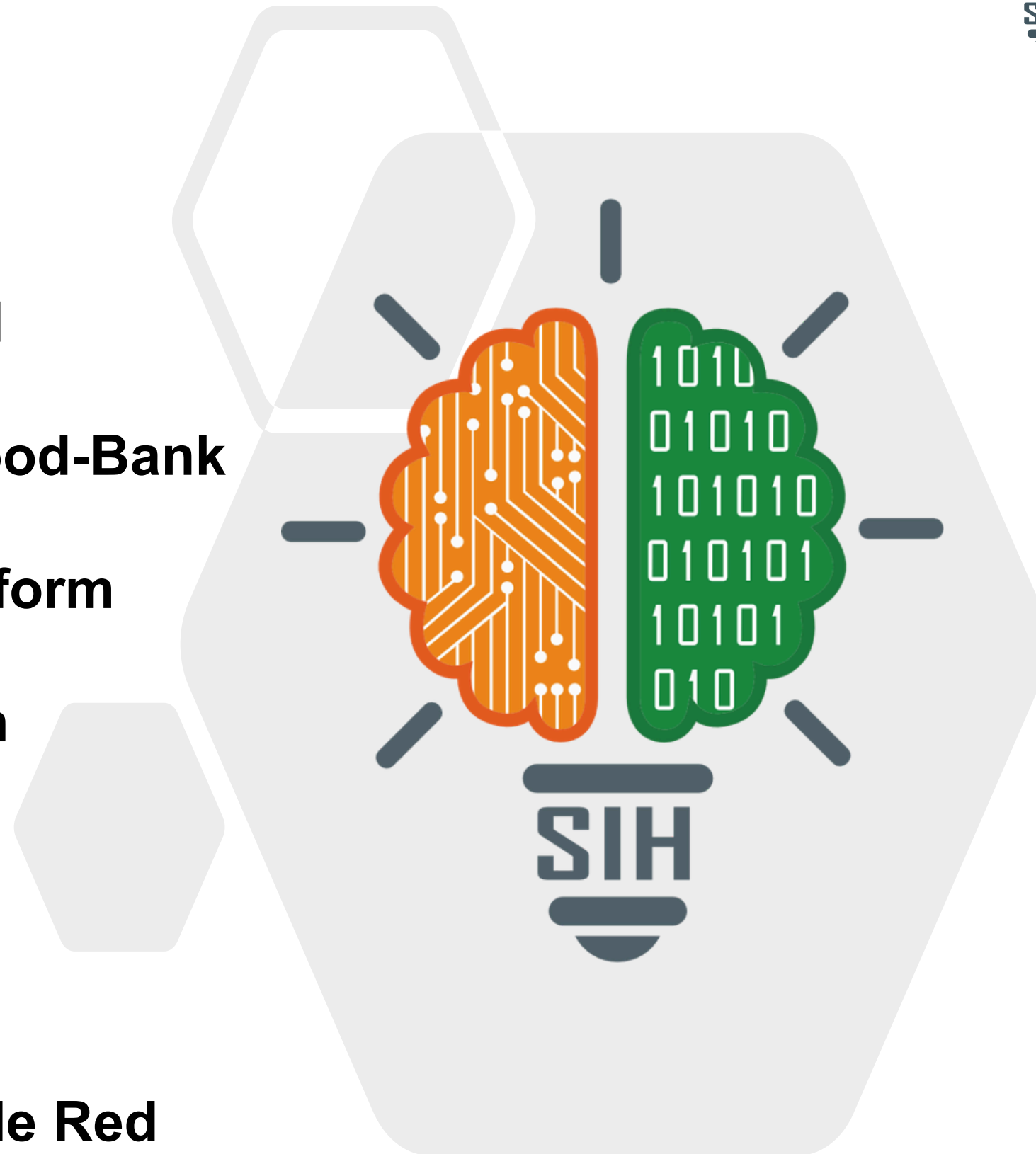
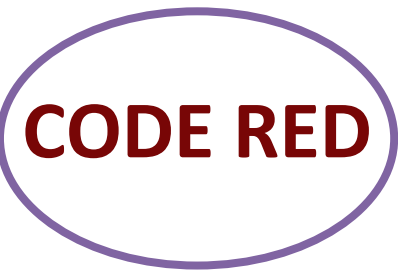


SMART INDIA HACKATHON 2026

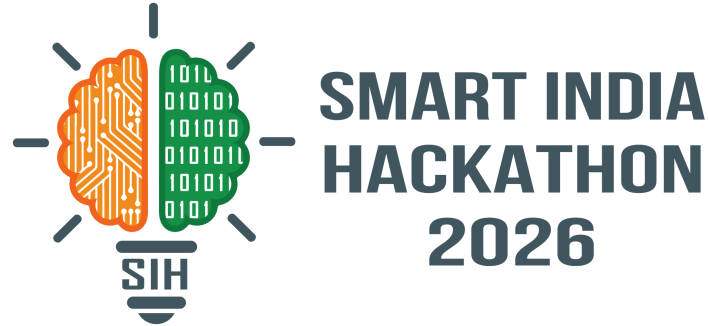


- **Problem Statement ID – SIH-2026-04-011**
- **Problem Statement Title- Smart Blood-Bank Inventory and Demand Forecasting Platform**
- **Theme- MedTech / BioTech / HealthTech**
- **PS Category- Software**
- **Team ID-**
- **Team Name (Registered on portal) - Code Red**

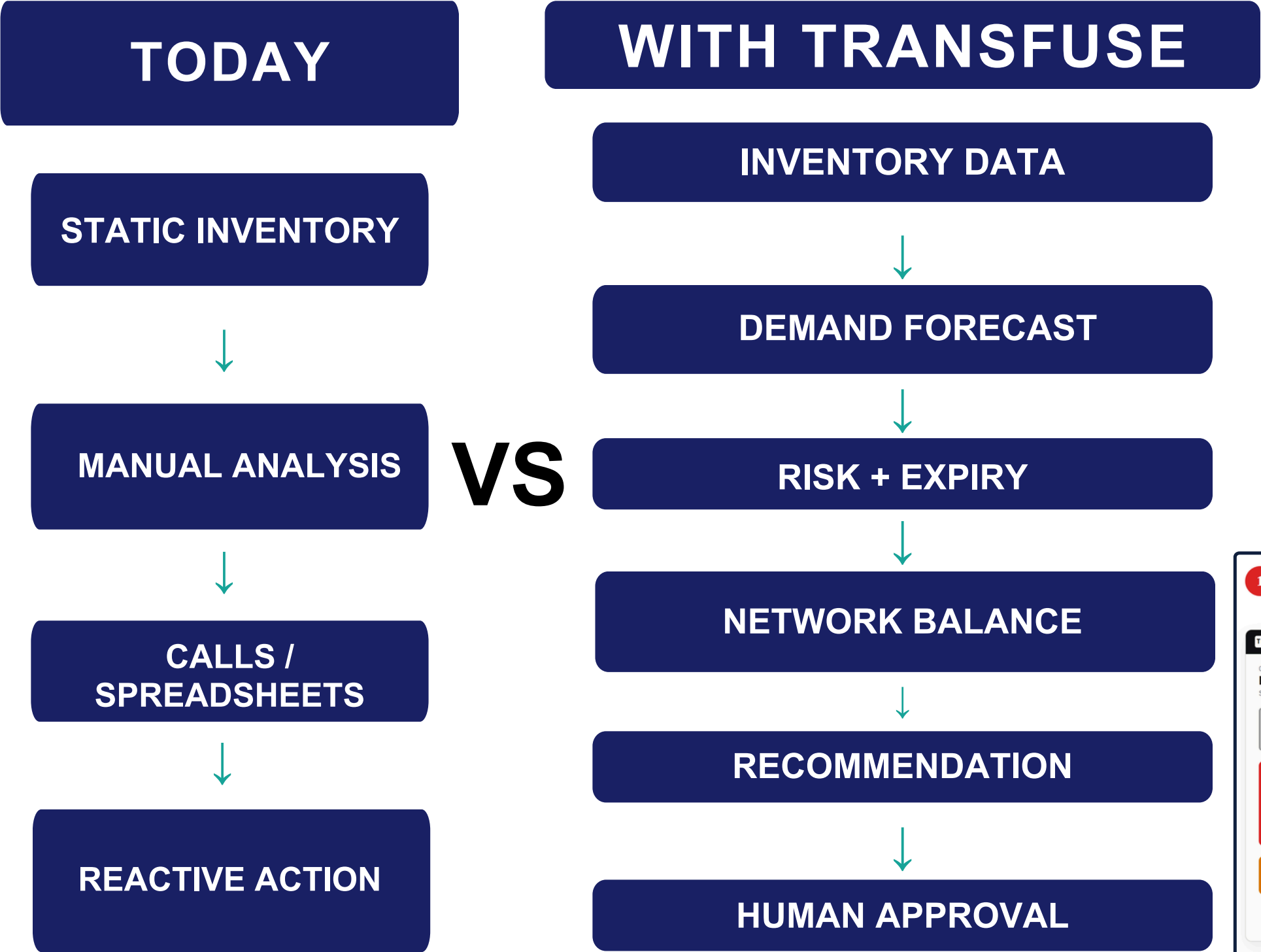




TRANSFUSE

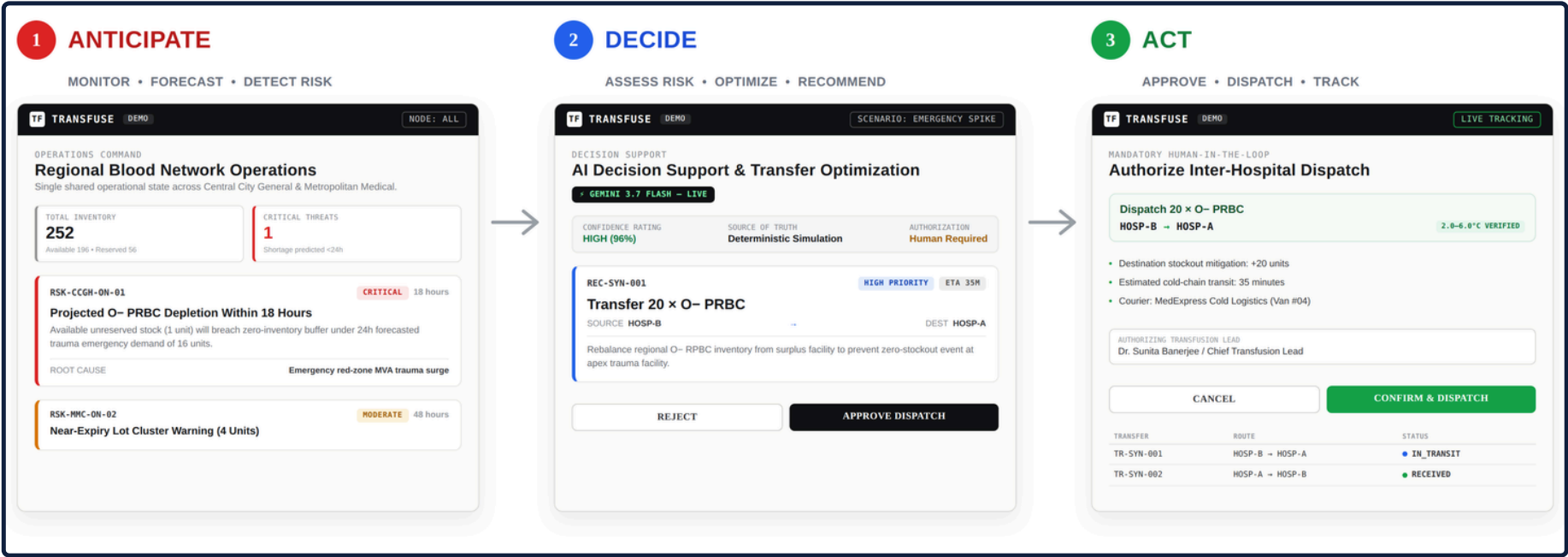
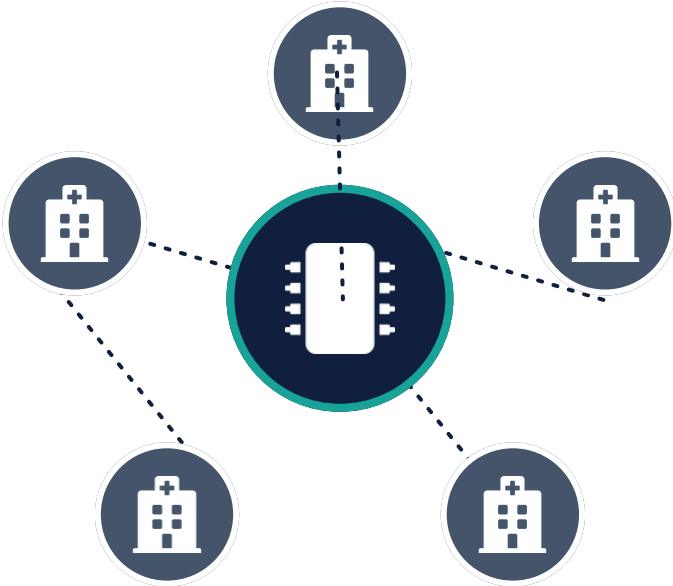


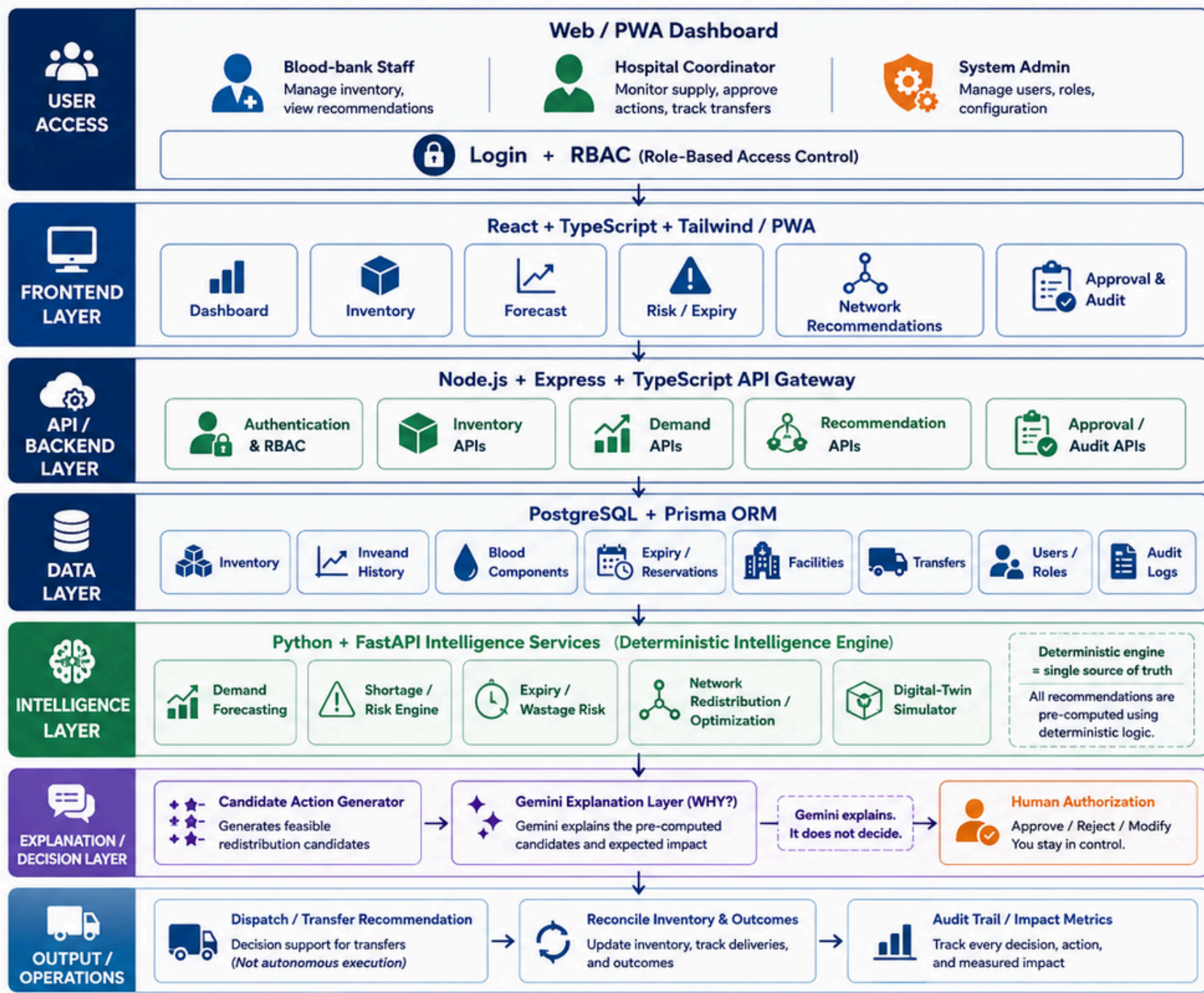
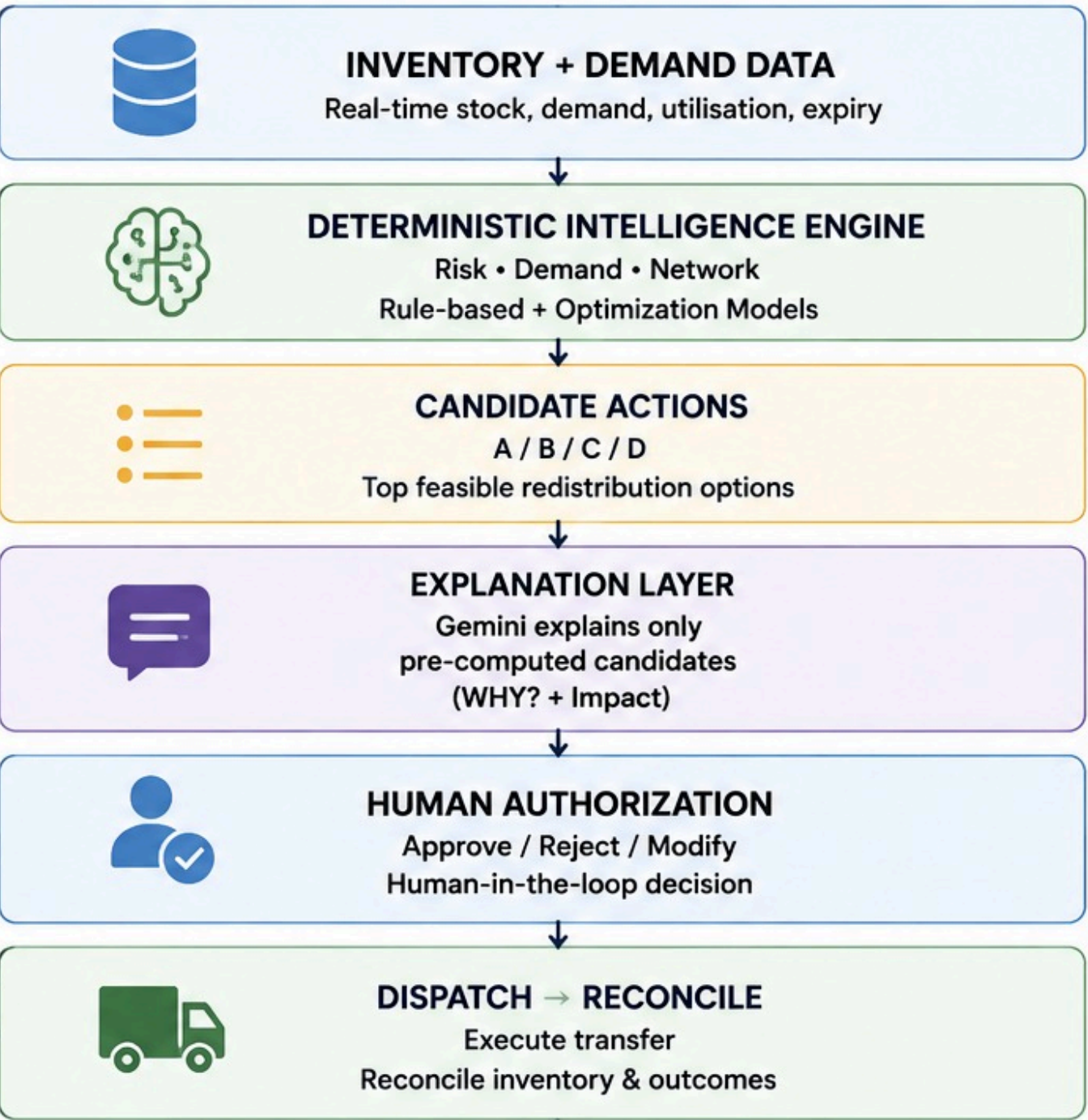
PROPOSED SOLUTION : TRANSFUSE is a predictive, network-aware decision-support platform that transforms blood-bank inventory data into actionable intelligence — forecasting demand, detecting shortage and expiry risks, identifying transferable surplus, and recommending explainable redistribution actions for human approval.



DIGITAL-TWIN SIMULATION NETWORK

5 Facilities × 8 Blood Groups × Multiple Components

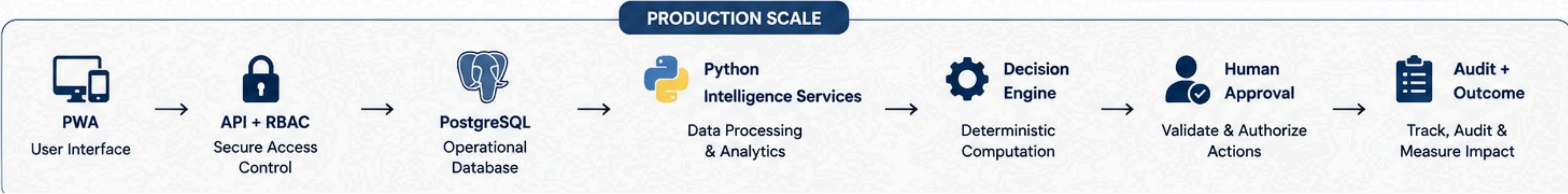




**GEMINI EXPLAINS.
IT DOES NOT DECIDE OR
ACT.**

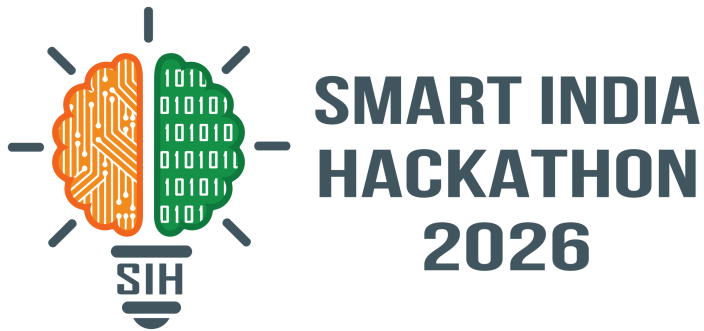
The deterministic engine is the source of truth. Gemini explains only pre-computed recommendations.

- WHY THIS DESIGN?**
- ✓ **Deterministic Engine**
Predictable Decisions
 - ✓ **LLM Explainability**
Understandable Reasoning
 - ✓ **Human Approval**
Controlled Execution
 - ✓ **Audit Trail**
Accountability & Trust





FEASIBILITY AND VIABILITY

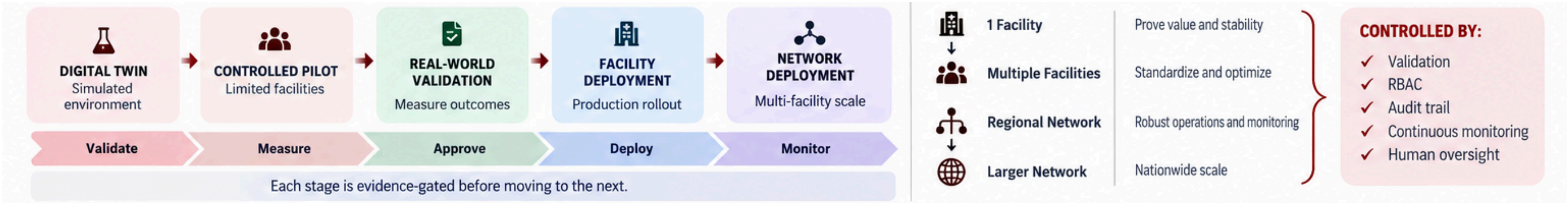


FEASIBILITY OF THE IDEA

DATA READY	INTELLIGENCE LAYER	HUMAN WORKFLOW	MODULAR SCALE	FUTURE SCOPE
• Inventory • Demand • Expiry • Facility Status • Network Data	• Forecast Demand • Detect Risks • Identify Transferable surplus • Generate Options	• Explain Recommendations • Review • Approve / Reject / Modify • Complete Audit Trial	• Facility Level • Multi-Facility • Regional Network • Network Scale	• Adaptive Intelligence • Real-time Network Optimization • Wider System Interoperability

SCALE CONTROL

Progressive expansion with built-in controls



DATA

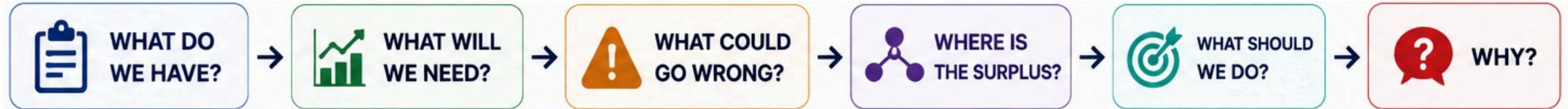
RISK : Incomplete or inconsistent data
MITIGATION: Automated data validation & quality checks

SAFETY

RISK : Incorrect or unsafe recommendation
MITIGATION: Human approval before every action

INTEGRATION

RISK : Compatibility with existing systems
MITIGATION: Modular API-based integration layer



IMPACT

BLOOD-BANK STAFF

- Earlier shortage warnings
- Expiry-risk visibility
- Explainable recommendations

HOSPITALS / COORDINATORS

- Better access to required stock
- Faster coordination
- Network-aware visibility

PATIENTS AND PUBLIC HEALTH

- Reduced risk of critical stock gaps
- Better use of donated blood
- Strong and reliable blood availability

BENEFITS

SOCIAL

- Reduced risk of stockout situations
- Improved equity in blood distribution
- Stronger community health systems

ECONOMIC

- Lower avoidable inventory loss
- Better utilization of existing resources
- Reduced operational inefficiencies

ENVIRONMENTAL

- Less expiry-driven blood wastage
- More efficient use of donated blood
- Reduced resource and energy waste



↓ STOCKOUT RISK

Earlier identification of projected shortages.



↓ EXPIRY / WASTAGE

Flags units approaching expiry before avoidable loss.



↓ RESPONSE DELAY

Structured recommendations reduce manual coordination.

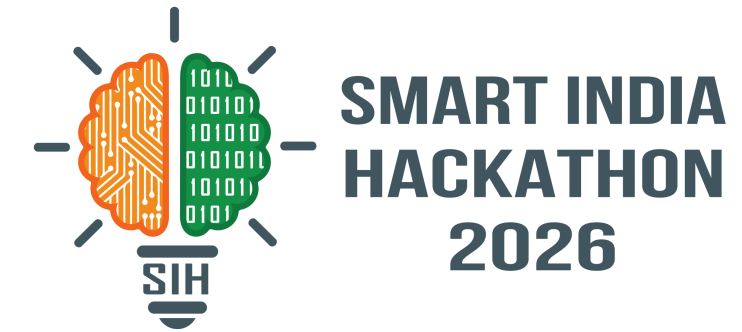


↑ RESOURCE UTILIZATION

Network visibility helps identify usable surplus.



RESEARCH AND REFERENCES



- **e-Raktkosh**

Ministry of Health & Family Welfare, Govt. of India

Centralized blood-bank management and inventory infrastructure.

Official website: <https://eraktkosh.mohfw.gov.in/>

- **Stanger et al. — 2012**

Blood Inventory Management: Hospital Best Practice

Transfusion Medicine Reviews. 2012;26(2):153–163.

DOI: 10.1016/j.tmr.2011.09.001

[PubMed — Stanger et al.](#)

- **Li et al. — 2022**

From demand forecasting to inventory ordering decisions for red blood cells.

Transfusion. 2022;62(1):87–99.

DOI: 10.1111/trf.16739

[PubMed — Li et al.](#)

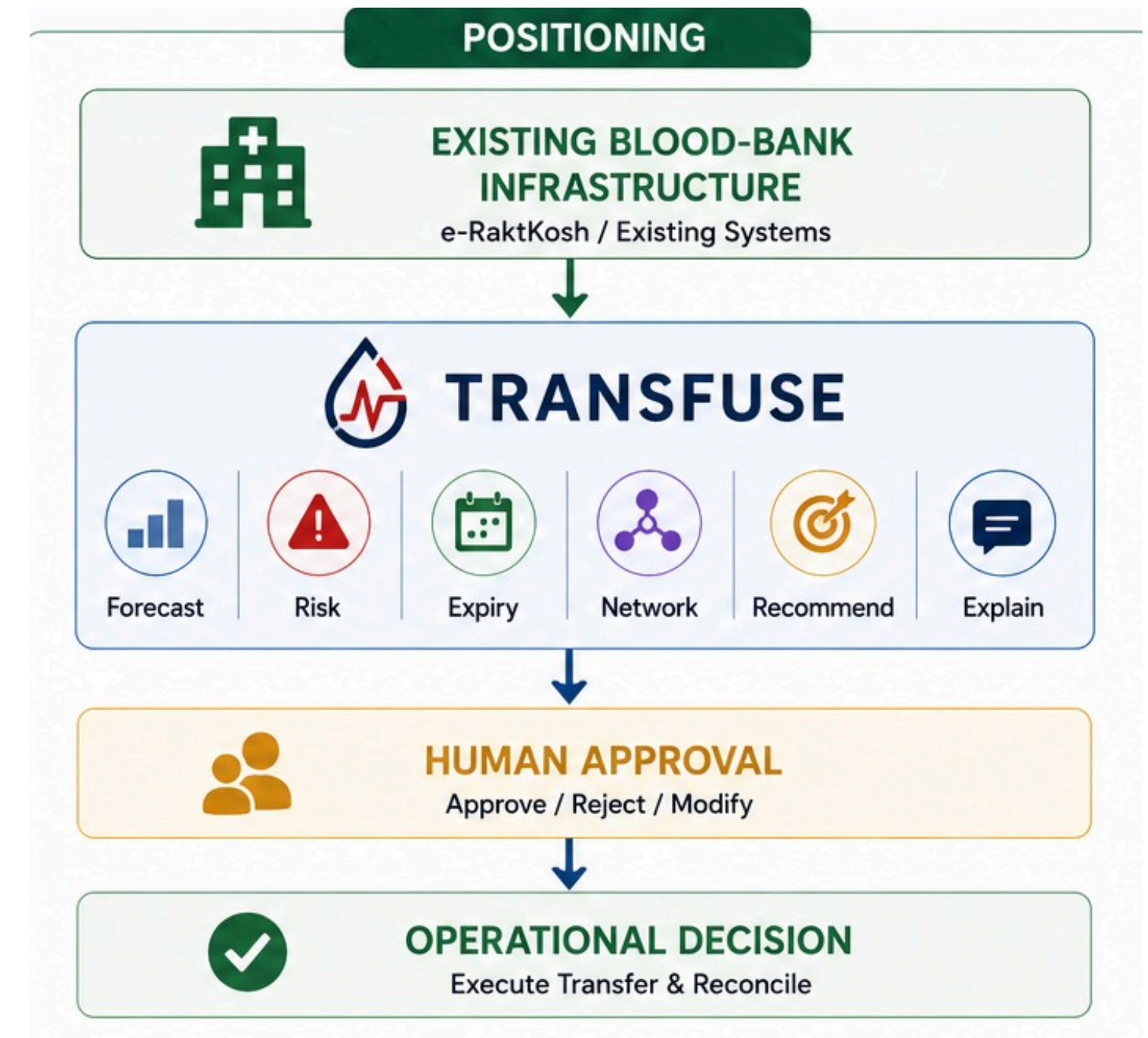
- **Engelke et al. — 2026**

AI and Blood Banking: Predicting Transfusion Demand — A Systematic Review

Transfusion Medicine and Hemotherapy, 2026.

DOI: 10.1159/000552969

[PubMed — 2026 Systematic Review](#)



TRANSFUSE ADDS PREDICTIVE + DECISION INTELLIGENCE WITHOUT REPLACING THE UNDERLYING BLOOD INFRASTRUCTURE.